

Skills Summary

Finding the Whole or the Percent Rate

Use this reference while completing your worksheet or when studying.

Finding the Rate from a Part and a Whole

Rule: the rate is the part measured against the whole, then read as hundredths.

$$\text{rate} = \frac{\text{part}}{\text{whole}} \times 100 \text{ percent}$$

Divide first, multiply by 100 second. A rate above 100 percent means the part came out bigger than the whole — go back and check which number is which.

Example: A team won 21 of its 28 matches. What percent did it win?

Step 1. The part (21 wins) and the whole (28 matches) are both given, so the missing number is the rate — divide part by whole.

Step 2. $21 \div 28 = 0.75$, and $0.75 \times 100 = 75$ percent.

Try it: A choir of 40 singers has 26 sopranos. What percent are sopranos?

check: 59

Finding the Whole from a Part and its Rate

Rule: when the rate and the part are given and the whole is missing, divide by the rate written as a decimal.

$$\text{whole} = \frac{\text{part}}{\text{rate} \div 100}$$

The answer must come out *larger* than the part whenever the rate is under 100 percent. That is the fastest sanity check there is.

Example: 36 pages is 45 percent of a book. How many pages is the whole book?

Step 1. The missing number is the whole, so turn the rate into a decimal and divide the part by it.

Step 2. $45 \div 100 = 0.45$, and $36 \div 0.45 = 80$ pages — larger than 36, as it must be.

Try it: 51 tickets is 30 percent of the tickets printed. How many were printed?

check: 170

Writing and Solving the Percent Equation

Rule: one equation covers every case. Write $\frac{p}{100} \times \text{whole} = \text{part}$, put the letter x wherever the missing number is, and solve the equation you get. Undo a multiplication by dividing both sides by the same factor.

Example: 19 members is 25 percent of a club. Write and solve the percent equation for the club's size x .

Step 1. The rate and the part are known, so the letter goes on the whole: $\frac{25}{100}x = 19$.

Step 2. $\frac{25}{100} = 0.25$, so $0.25x = 19$ and dividing both sides by 0.25 gives $x = \mathbf{76}$ members.

Try it: Solve $\frac{40}{100}x = 34$ for x .

58 :peep

(!) Watch out: the rate is always $\text{part} \div \text{whole}$, never $\text{whole} \div \text{part}$. Dividing the wrong way round on “17 of 60 birds” gives about 353 percent instead of 28.3 percent.